

영국의 행정 구역은 크게 Lieutenancy area인 여러 개의 County로 나뉜다. 잉글랜드의 경우, ceremonial county라고 불리며, 웨일즈는 preserved county, 스코틀랜드와 북아일랜드는 lieutenancy area라고 칭한다. 우리는 광역런던시(Greater London)에 한정하여 집값 및 전수조사 데이터를 분석할 예정이므로 잉글랜드에 국한해 영국의 행정 구역을 설명하겠다. 잉글랜드에는 총 48개의 ceremonial county가 존재하며, 각 county에는 왕정의 개인 대리인격인 lord-lieutenant와 사법적 대리인인 sheriff(주장관)이 한 명씩 배치된다. County는 non-metropolitan과 metropolitan으로 그 유형이 구분되는데, 우리나라의 상황에 맞춰 해석한다면 경기도와 인천광역시도 서로 다른 행정 구역인 점과 마찬가지로. Non-metropolitan county는 또 다시 여러 개의 non-metropolitan district로 나뉜다. 이를 ‘이중 자치 단체 의회 구조(two-tier council arrangement)’라고 부른다. 마치 경기도에 여러 개의 시가 존재하고, 경기도의회와 각 시의회가 따로 존재하는 것과 마찬가지다. 다만, 어떤 county는 여러 개의 district로 행정 업무가 분산되지 않고 여러 마을(settlement)을 county council 차원에서 총괄하기도 하는데, 이를 ‘unitary authority areas’라고 부른다. 비록 아래 표에서는 metropolitan county와 광역 런던시의 의회 구조를 각각 metropolitan borough, London borough 등으로 다르게 표기하기는 하지만, 실질적으로 그 둘도 unitary authority area로 볼 수 있다. 달리 말해, borough는 우리나라 행정 구역상 ‘구’에 해당한다.

광역런던시는 총 32개의 London boroughs와 별도의 행정 자치구역인 City of London으로 구성된다. 또 다시 우리나라의 입장에서 생각해 본다면, 서울특별시 중구만 그 역사성 및 행정, 경제적 중요성을 고려해 별도의 자치 단체를 설립할 수 있도록 허가했을 때 서울특별시가 24개의 자치구와 하나의 중구로 이루어지는 것과 마찬가지다. 32개의 London boroughs 중 12개는 Inner London, 나머지는 Outer London boroughs로 분류된다. City of London을 포함한 모든 광역런던시는 각각의 자치 의회(local council)뿐만 아니라 런던 시장이 그 수장인 런던광역행정청(Greater London Authority; GLA)에 의해서도 운영된다. 여러 행정 업무 중 보행자 교통, 경찰, 소방 업무는 GLA가 담당하고, 주택, 전략 및 교통 계획 수립, 도로 관리 등의 업무는 GLA와 각 London borough의 의회가 같이 담당하며, 나머지 업무는 모두 각 구의회가 담당한다. 각 London borough는 또 다시 몇 개의 지방 자치단체 선거구(ward)로 나뉜다. City of London의 경우 총 25개의 ward로 이루어지며, 비단 광역런던 시의원뿐 아니라 일종의 부시장격인 alderman도 각 ward마다 한 명씩 선출한다. City of London을 제외한 다른 32개의 London boroughs들은 총 679개의 ward로 나뉜다. 따라서 광역런던시 전체로 보았을 때는 총 701개의 ward가 존재한다. 여기서 한 가지 주의해야 하는 점은, 지방 자치단체 선거구인 ward와 국회 선거구(parliamentary constituency)는 다르게 구획된다. 앞서 말했듯 광역런던시에는 701개의 ward가 있지만, 국회 의원(parliament; 앞서 언급한 의회(council)과 다르게 하기 위해 국회로 번역함.) 선거를 위해서는 73개의 선거구로 구분된다. 각 ward는 오직 하나의 borough에만 속할 수 있는 반면, 선거구는 borough에 한정하지 않고 여러 개의 ward를 합쳐서 구성하므로 하나의 선거구가 복수의 borough에 걸쳐 있을 수도 있다. 하나의 borough는 대체로 2, 3개의 국회 선거구로 이루어진다.

Following [Brexit](#) the UK [ONS](#) replaced the [EU NUTS1](#) regional model with its own [International Territorial Level](#) model, continuing the treatment of the 3 [Home Nations](#) alongside the 9 [Regions of England](#).

Overview of administrative divisions of the UK

Country	United Kingdom ^[L]						
Part of UK	England						Wales ^[L]
Authority area	Greater London Authority and combined authority areas, ^[C] other areas						Scotland ^[L]
Lieutenancy area	Ceremonial counties						Northern Ireland ^[L]
County type	Non-metropolitan	Metropolitan	Unique				Preserved counties
County council area	Two-tier council area ^[C]	Unitary authority areas ^[C] ^[U]	None	None	City of London ^[C]	Isles of Scilly ^[C] ^[S]	Lieutenancy areas
District	Non-metropolitan districts ^[B] ^[C]	Metropolitan boroughs ^[B] ^[C]	London boroughs ^[C]				Lieutenancy areas
Parish	Civil parishes ^[C] ^[T]				None	Civil parishes	Communities ^[M]
							Communities ^[V]
							(Civil parishes) ^[W]

Notes

Service	Greater London Authority	London borough councils
Education		✓
Housing	✓	✓
Planning applications		✓
Strategic planning	✓	✓
Transport planning	✓	✓
Passenger transport	✓	
Highways	✓	✓
Police	✓	
Fire	✓	
Social services		✓
Libraries		✓
Leisure and recreation		✓
Waste collection		✓
Waste disposal		✓
Environmental health		✓
Revenue collection		✓

Price-related variables:

```
> str_subset(colnames(london_2011_supra), "Price|price")
[1] "rentEstimate_lowerPrice" "rentEstimate_currentPrice" "rentEstimate_upperPrice" "saleEstimate_lowerPrice"
[5] "saleEstimate_currentPrice" "saleEstimate_upperPrice" "history_price"
> |
```

- rentEstimate : current price estimate for rents
- saleEstimate: current price estimate for sales
- history_price: history price for sales (specific dates provided)

```
> london_2011_supra |> group_by(fullAddress) |> summarise(n = n()) |> filter(n >= 50) |> arrange(desc(n))
# A tibble: 17 x 2
```

fullAddress	n
<chr>	<int>
1 74 Western Beach Apartments, 36 Hanover Avenue, London, E16 1DZ	224
2 Oak Apple Court, Gables Close, London, SE12 0UB	150
3 Horley Court, 46 Inverness Terrace, London, W2 3JA	136
4 Aurora Point, 3 Winchester Square, London, SE8 3FQ	104
5 6, 14-16 Church Hill, London, E17 3AG	102
6 Chedworth House, 227 West Green Road, Tottenham, London, N15 5EH	92
7 Clarendon Court, 2 Brackley Street, London, EC1Y 0AB	84
8 Elm Court, Bromley Road, London, SE6 2TX	72
9 4, 4A Maida Vale, London, W9 1SP	64
10 Highgate Heights, 77 Shepherds Hill, Hornsey, London, N6 5RF	63
11 The Lindales, Grasmere Road, Tottenham, London, N17 0HE	62
12 Beaufort House, Talbot Road, Tottenham, London, N15 4DR	58
13 Selkirk Court, Whitley Road, Tottenham, London, N17 6RF	58
14 Melior Court, 79 Shepherds Hill, Hornsey, London, N6 5RQ	54
15 Holmhurst, Beacon Road, Hither Green, London, SE13 6EG	52
16 Kings Parade, High Road, Tottenham, London, N17 6QL	52
17 Panorama Court, 56 Shepherds Hill, Hornsey, London, N6 5RP	50

- 17 particular houses with more than 50 price histories.

Longitude, Latitude (with capital letters): from `postal\_codes` dataset

Longitude, latitude (non-capitalised letters): from `house\_prices` dataset

```
> london_2011_supra |> filter(round(abs(latitude - Latitude), 4) != 0 | round(abs(longitude - Longitude), 4) != 0) |> select
(latitude, Latitude, longitude, Longitude)
```

```
# A tibble: 269,517 x 4
```

	latitude	Latitude	longitude	Longitude
	<dbl>	<dbl>	<dbl>	<dbl>
1	51.5	51.5	-0.0922	-0.0927
2	51.6	51.6	-0.140	-0.139
3	51.5	51.5	0.00697	0.00704
4	51.5	51.5	-0.00757	-0.00745
5	51.6	51.6	-0.00218	-0.00220
6	51.6	51.6	-0.140	-0.139
7	51.6	51.6	-0.0202	-0.0210
8	51.6	51.6	-0.0202	-0.0210
9	51.5	51.5	-0.341	-0.340
10	51.5	51.5	-0.189	-0.189

```
# i 269,507 more rows
```

```
# i Use `print()` to see more rows
```

- (above) 소수점 넷째자리까지 반올림했을 때 postal_codes 위치 정보와 house_prices 위치 정보가 일치하지 않는 경우: 전체 418,201건 중 269,507건.
- 이 경우를 제외하고 보아야 할까? 그럴 필요는 없어 보임.

POSTAL_CODE

Geographical Variables ~

- (Continuous) **“Longitude”** / **“Latitude”** / **“Altitude”** / “Easting” / “Northing” / “Grid Ref” / “Distance to sea” [7]
- (Discrete; Administrative) **“Ward”** = **“Ward Code”** < “District” = “District Code” < “Country”; “Parish” / “Parish Code” [7]
- (Discrete; Postal) **“Postcode”** < **“Postcode district”** < “Postcode area”; “Plus Codes” [4]
- (Discrete; ONS Standard) “ITL level 3” < “ITL level 2” [2]
- (Discrete; ONS Decadal London Census Standard) “Census output area 2021” < **“Lower layer super output area 2021”** = **“LSOA21 Code”** < “Middle layer super output area 2021” = “MSOA21 Code”; “Census output area” < **“Lower layer super output area”** = **“LSOA Code”** < “Middle layer super output area” = “MSOA Code” < “Local authority” < “County” = “County Code” [13]
- (Discrete; Constitutional) **“Constituency Code 2024”** / **“Constituency Name 2024”** [2]

Property-related Variables ~

- (Property itself) “UPRNs” / “Roads” / **“In Use?”** / **“Property Type”** [4]
- (Neighbourhood) **“Rural/urban”** / “Population” [C] / “Households” [C] / **“Police force”** / **“Water Company”** / **“Average income”** [C] / **“Sewage company”** / **“IMD Decile”** / “Index of Multiple Deprivation” [C] / “Quality” (of IMD) [10]
- (Transportation) **“London zone”** / **“Nearest Station”** / **“Distance to station”** [C] / **“Travel to work area” (Destination)** [4]

Uncertain & Unnecessary Variables ~ [7]

- (Date) “Introduced” / “Terminated”
 - 아마도 데이터셋에 처음 포함된 날짜?
- “Built up area” / “Region” / “National Park” / “User Type”
- “Last updated” (the date when the postal code csv is last updated)

References:

- “Easting” / “Northing”
 - Eastings are the vertical lines running from the top to bottom and divide the map from west to east. Northings are the horizontal lines running left to right and divide the map from north to south. Each easting and northing is numbered on the map margin at 1 km intervals; numbers are used to find a grid reference.
- “Parish”
 - In England, an unparished area is an area that is not covered by a civil parish. Most urbanised districts of England are either entirely or partly **unparished**. Many towns and some cities in otherwise rural districts are also unparished areas and therefore no longer have a town council or city council, and are instead directly managed by a higher local authority such as a district or county council.
- “Index of Multiple Deprivation” [cont.]
 - The Index of Multiple Deprivation (IMD) datasets are small area measures of relative deprivation across each of the constituent nations of the United Kingdom. Areas are ranked from the most deprived area (rank 1) to the least deprived area. Each nation publishes its data on its own data portal. Each nation measures deprivation in a slightly different way but the broad themes include *income, employment, education, health, crime, barriers to housing and services, and the living environment*.
- “Plus code”
 - **Plus Codes are designed to be used like street addresses** and may be especially useful in places where there is no formal system to identify buildings, such as ...
- “UPRNs”
 - The Unique Property Reference Number (**UPRN**) is a unique numeric identifier for every spatial address in Great **Britain**. It is created by local authorities ...
- Built up area

- The Greater London Built-up Area, or Greater London Urban Area, is a [conurbation](#) in south-east [England](#) that constitutes the continuous [urban sprawl](#) of [London](#), and includes surrounding adjacent urban towns as defined by the [Office for National Statistics](#).^[1] It is the largest urban area in the [United Kingdom](#) with a population of 9,787,426 in 2011.^[1]
- MSOA Code
 - Hierarchy: Census Output area < LSOA < MSOA < Local Authority < County (Greater London or City of London)
 - LSOA: intermediate in size between census Output Areas (OAs) and local authorities.
 - <https://gis.stackexchange.com/questions/218387/understanding-lsoa-and-msoa-boundaries-uk>
 - [**https://www.ons.gov.uk/methodology/geography/ukgeographies/censusgeographies/census2021geographies**](https://www.ons.gov.uk/methodology/geography/ukgeographies/censusgeographies/census2021geographies)

HOSUE_PRICES

Geographical Variables~

- (Postal) "fullAddress" / "postcode" / "country" / "outcode"
- (Continuous) "latitude" / "longitude"

Property-related Variables~

- (Property itself) "bathrooms" / "bedrooms" / "floorAreaSqM" / "livingRooms" / "propertyType" / "currentEnergyRating"
- (About its residents) "tenure"
- (About its prices) "rentEstimate_lowerPrice" / "rentEstimate_currentPrice" / "rentEstimate_upperPrice" / "saleEstimate_lowerPrice" / "saleEstimate_currentPrice" / "saleEstimate_upperPrice" / "saleEstimate_confidenceLevel" / "saleEstimate_ingestedAt" / "saleEstimate_valueChange.numericChange" / "saleEstimate_valueChange.percentageChange" / "saleEstimate_valueChange.saleDate" / "history_date" / "history_price" / "history_percentageChange" / "history_numericChange"

Unnecessary Variables~

- "country" / "latitude" / "longitude" / "outcode" / "propertyType" / all "lowerPrice", "upperPrice" / "saleEstimate_valueChange.numericChange"
- (postal_code) dataset's "Property Type" seems more informative than "propertyType" from (house_prices) as the latter breaks down the house types too narrowly.
- "numericChange" redundant of "percentageChange"

JOIN Keys

- "Postcode"

References:

- "outcode" = "Postcode district" from (postal_code) dataset.
- "tenure"
 - Housing tenure is a **financial arrangement and ownership structure** under which someone has the right to live in a house or apartment.
 - 2 categories: "freehold" / "leasehold"
- "saleEstimate_confidenceLevel"
 - Confidence in the sale price estimate (LOW, MEDIUM, HIGH).
- Property Types: <https://www.tembomoney.com/learn/what-are-the-different-types-of-houses-uk>
 - 7 common types in the UK:



- "saleEstimate_valueChange..."
 - "ingestedAt" = the date when the "saleEstimate_currentPrice" is appraised.
 - "valueChange.saleDate" = the date when the previous "saleEstimate_currentPrice" was appraised.
 - "valueChange.numericChange" = the difference between the estimated sales price of the same property back then(valueChange.saleDate) and now(ingestedAt).
 - Hence, we can calculate back the past estimated sales price of the properties.